

Computer Peripherals - Arduino Lab 1

1. Introduction to Arduino Board

First of all please watch [this](#) brief video introduction about the Arduino boards generally.

2. Introduction to Arduino IDE

To program an Arduino board you need the development environment for it. You can download the Arduino IDE from [here](#). For a comprehensive tutorial about the whole Arduino programming, check [this link](#).

Very briefly, the Arduino program structure contains the following two main parts: *setup* and *loop*.

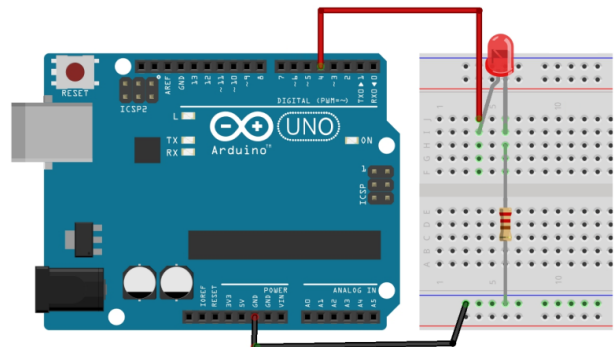
The *setup* function is called when a sketch starts. Use it to initialize the variables, pin modes (input/output), start using libraries, etc. The *setup* function will only run once, after each power up or reset of the Arduino board.

The *loop* function does precisely what its name suggests, and loops consecutively, allowing your program to change and respond. Use it to actively control the Arduino board.

3. Led Blinking Example

In the *setup* section we can use the *pinMode* function to set the digital pin 4 as an output. In the *loop* section we can use the *digitalWrite* function to set the output *high*. By using the *delay* function we can add a 500 milliseconds wait to our program, then we can set

```
void setup() {  
  pinMode(7, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(7, HIGH);  
  delay(500);  
  digitalWrite(7, LOW);  
  delay(500);  
}
```



our output to *low* with the same *digitalWrite* function, and again we can add another half-second to wait. And that's all folks, we are ready to upload our first program tho the physical Arduino board and test it, but first please check also the connection scheme in the figure.

3.1. The task



Please create a physical setup for three different colored led connected to the Arduino board (preferable red, yellow and green). Then write a code that simulates the behaviour of a traffic semaphore like in [this video](#).

Bibliography

This laboratory guide is based on [DroneHow](#) and [HackerEarth](#) Youtube channels contents, for more interesting projects checks those channels.